Tanav Ravella

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Summary

Mathematics & Computer Science student at UIUC. Builds simulation and data software in Python and C++ for a 50-member engineering team, and built the end-to-end forecasting pipeline for a national risk-modeling competition. Prior healthcare research internship at the University of Chicago working with cancer imaging data.

Education

University of Illinois Urbana-Champaign — Champaign, IL

B.S. Mathematics & Computer Science — *Aug. 2026 – May 2029 (Expected)*

- **Relevant Coursework:** CS 128 (Computer Science II, OOP and design in C++), CS 173 (Discrete Structures), MATH 314, Linear Algebra, Calculus III, STAT 400 (Probability & Statistics, Spring 2027); placed out of CS 124 (Intro to CS I, Java) by proficiency examination
- **Activities:** Illini Solar Car (Strategy & Modeling); SIG ICPC; SIGma; Project Code

Projects

Race-Strategy Simulator, Illini Solar Car | *Python, NumPy* — Fall 2026 – Present

- Build battery state-of-charge models in Python on the Strategy & Modeling subteam, whose simulator informs the race crew's speed and energy decisions
- Model vehicle physics and cornering losses to find the optimal line and speed profile around a track

Wildfire Risk Model, Modeling the Future Challenge | *Python, pandas, scikit-learn* — 2024 – 2026

- Reached Semifinalist in the Actuarial Foundation's national risk-modeling competition, building the wildfire forecasting pipeline end to end
- Merged four decades of records with pandas and benchmarked four model families, selecting an ARIMA forecast that beat regression, SVM, and random-forest baselines

Experience

Research Intern — Summer 2025

University of Chicago SPARK Program — *Chicago, IL*

- Quantified how pigment composition across ∼30 tattoo inks distorts cancer radiology imaging, collaborating with senior researchers on the imaging analysis
- Presented findings on diagnostic accuracy to the wider research group

Captain — 2024 – 2026

Stevenson Math Team — *Lincolnshire, IL*

- Led a 100+ member team to 1st or 2nd place at the ICTM State Contest and North Suburban Math League every year from 2023–2026, running weekly sessions on algorithmic problem-solving and discrete math
- Built and maintained the team's shared review library, documenting multi-step problems as written walkthroughs newer members could follow without help

Honors & Awards

- USACO Silver Division | 4× AIME Qualifier (2022–2025) | AMC 10A (top 2.5% nationally) | USAPhO Honorable Mention (2026)

Technical Skills

Languages: C++, Python, Java, SQL, HTML, CSS
Machine Learning & Data: scikit-learn, pandas, Polars, NumPy
AI & Development Tools: Claude Code, Git/GitHub, Jupyter/Google Colab, LaTeX
Concepts: Machine Learning, Data Structures & Algorithms, Object-Oriented Programming & Design, Relational Databases, Statistical Modeling, Telemetry & Metrics